

Subject: Biology and Environmental Science (CY202)

Question Bank

Unit 1: Cells and Biomolecules - Structure

Short questions:

1. Draw a labelled diagram of a plant cell.
2. What is a peptide bond?
3. What is a glycosidic bond?
4. Give example of acidic/basic/neutral/sulphur containing/aromatic amino acid. Mention respective three letter and single letter code with structure.
5. Differentiate between saturated and unsaturated lipid.
6. What are suicide bags of cells? Why are they termed so?
7. Why mitochondria is known as the power house of the cell?
8. What are glycoprotein and lipoprotein?
9. Mention two examples disaccharide with their constituent monosaccharides.
10. Write down any two functions of carbohydrates/lipid/protein.
11. What do you mean by structural and functional proteins.
12. Define ribosome. Write the functions of ribosome.
13. Draw the bilayer structure of a lipid.
14. Write down two differences between rough endoplasmic reticulum and smooth endoplasmic reticulum.
15. What kind of interactions are responsible for stabilization of secondary structure of protein.
16. Write down two important functions of mitochondria.
17. What are enzymes? Give two examples of enzymes that are found in body.
18. Define chlorophylls. Write down two functions of chlorophylls.
19. Write down two functions of carotenoids.

Broad questions:

1. Describe the fundamental characteristics of living organisms.
2. Elaborate on different types of organic constituents present in the living cells with examples.
3. Compare and contrast between prokaryotic and eukaryotic cells.
4. Describe the functions of cell membrane, cell wall and cytoplasm.
5. Describe the structure and functions of cell organelles such as the endoplasmic reticulum, Golgi apparatus, mitochondria, and lysosomes.
6. What are the differences between the plant cells and animal cells?
7. Write down the differences between the animal cells and bacterial cells?
8. Define cell. Write down five characteristics of cells. Draw a structure of cell with labelled diagram.

4 - 5

9. What are enzymes? Discuss their structure, characteristics and working functions in the body.
10. Distinguish between primary, secondary tertiary and quaternary structures of protein.
11. Depict the primary structure of the following peptide code: FYP (say), ATF, DCP
12. Classify animal tissues with their functions.